

B.Tech. Degree VII Semester Special Supplementary Examination ***April 2016***

EB/EC 1705 E4 MECHATRONICS

(2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A

(Answer *ALL* questions)

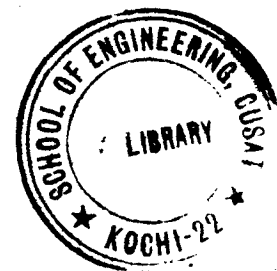
(8 × 5 = 40)

- I. (a) What is Mechatronics? Discuss on the scope of Mechatronics.
- (b) Describe the input output relationship for rotational translation system.
- (c) What is the need for bearings? Explain the different types of ball bearings.
- (d) Explain the different types of tachometers.
- (e) Explain the working of robotic system.
- (f) What are the classifications of numerical control systems?
- (g) Discuss on the importance of MEMS in mechatronics.
- (h) What is derivative control Explain how PD control is implemented using op-amp circuit?

PART B

(4 × 15 = 60)

- II. (a) Explain the following performance measures for a second order system. (10)
 - (i) Rise time.
 - (ii) Peak time.
 - (iii) Overshoot.
 - (iv) Settling time.
- (b) Explain the input output relation of the rack and pinion. (5)
- OR**
- III. (a) Obtain the relationship for the basic building blocks of mechanical and electrical systems. (10)
- (b) Write short notes on dynamic response of systems. (5)
- IV. Explain the following with neat diagram. (15)
 - (i) Rotary actuators.
 - (ii) Ratchet and pawl.
 - (iii) Operation of CAM.
- OR**
- V. Explain the following. (15)
 - (i) Stepper motor.
 - (ii) Encoders.
 - (iii) Pressure control valves.



(P.T.O.)

- VI. (a) Explain a CNC system with a neat block diagram. (10)
(b) List the design consideration of NC machine tool. (5)

OR

- VII. (a) Explain the classification of robots. What are the importance of the degrees of freedom of a robot? Explain with example. (8)
(b) Explain the classification of NC system. (7)

- VIII. (a) Explain the structure of PLC and mention the selection criteria for the same with a neat diagram. (10)
(b) List the advantages and applications of MEMS. (5)

OR

- IX. Explain the various closed loop control modes in detail. (15)